

ADELBODEN, Switzerland

WMO: 067350

Lat: 46.4917N

Lon: 7.5607E

Elev: 1328

StdP: 86.35

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.3	-11.2	-18.3	0.9	-10.2	-16.0	1.1	-8.0	10.8	5.4	9.2	4.8	1.0	190	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	24.9	16.3	23.3	15.8	21.6	15.1	17.4	22.9	16.5	21.7	15.7	20.5	3.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.7	19.8	14.4	12.1	19.1	13.7	11.5	18.3	54.3	23.3	51.5	21.8	48.9	20.6	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.4	6.0	5.0														
(5)			DB	-15.6	27.7	2.4	1.4	-17.3	28.7	-18.7	29.6	-20.1	30.4	-21.8	31.4	
			WB	-16.2	19.3	2.3	0.9	-17.9	19.9	-19.2	20.4	-20.5	20.9	-22.1	21.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-1.3	-1.3	2.1	6.0	9.2	13.6	15.1	14.6	11.2	8.3	3.1	-0.1
	DBStd	7.28	4.73	5.23	4.58	4.17	4.06	3.97	3.39	3.55	3.57	4.21	4.74	4.81
	HDD10.0	1751	349	317	245	129	64	12	3	4	29	79	208	312
	HDD18.3	4247	607	550	503	369	282	149	109	124	215	311	456	571
	CDD10.0	572	0	0	1	10	40	119	161	147	64	27	2	1
	CDD18.3	24	0	0	0	0	0	7	9	8	0	0	0	0
	CDH23.3	136	0	0	0	0	1	42	50	42	1	0	0	0
	CDH26.7	11	0	0	0	0	0	4	4	2	0	0	0	0
(14) Wind	WSAvg	1.7	1.6	1.7	1.9	1.8	1.9	1.8	1.7	1.6	1.5	1.4	1.5	1.7
(15) Precipitation	PrecAvg	994	71	70	66	67	97	98	109	105	71	70	81	88
	PrecMax	1273	211	248	162	134	183	187	202	195	141	154	210	232
	PrecMin	742	8	12	20	17	36	51	45	34	34	3	12	3
	PrecStd	131	49	56	38	28	45	32	35	37	28	36	47	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.7	14.2	19.1	22.6	27.1	27.0	26.9	22.0	19.9	14.8	12.4
		MCWB	3.7	3.9	6.3	9.6	13.4	16.3	17.0	16.5	14.5	10.2	7.0	4.7
	2%	DB	8.5	9.0	12.2	16.8	20.0	24.4	24.7	24.4	20.2	17.7	13.0	9.6
		MCWB	3.0	2.8	5.3	8.9	12.3	16.1	16.1	16.4	14.2	10.6	6.4	3.4
	5%	DB	6.7	7.0	10.5	14.8	18.1	22.4	22.9	22.5	18.5	15.6	11.2	7.6
		MCWB	2.7	2.2	4.5	8.0	11.2	15.3	16.0	15.9	13.3	9.6	5.9	3.0
	10%	DB	5.2	5.3	8.6	12.8	16.0	20.4	21.1	20.6	16.6	13.8	9.4	6.1
		MCWB	1.8	1.3	3.6	6.9	10.3	14.5	15.3	15.2	12.3	9.2	5.3	2.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	5.2	7.6	10.8	14.9	18.1	18.5	18.5	16.1	12.9	8.6	5.7
		MCDB	8.5	9.4	12.1	17.0	20.6	24.4	24.3	23.8	20.1	16.8	11.7	9.7
	2%	WB	4.3	3.8	6.1	9.5	13.2	16.9	17.3	17.2	14.8	11.6	7.5	4.7
		MCDB	7.1	7.1	10.6	15.2	18.1	22.8	22.7	22.5	18.9	15.1	11.0	8.1
	5%	WB	3.2	2.7	5.1	8.5	11.9	15.9	16.4	16.2	13.8	10.7	6.6	3.7
		MCDB	6.2	6.3	9.4	13.7	16.7	21.1	21.5	21.1	17.4	14.2	10.3	6.9
	10%	WB	1.9	1.6	4.1	7.4	10.8	14.9	15.6	15.4	12.8	9.7	5.6	2.6
		MCDB	4.8	5.1	7.7	12.1	15.2	19.6	20.2	19.8	15.8	12.9	9.1	5.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.2	7.8	8.7	8.6	8.7	8.8	8.4	7.9	7.8	6.6	6.6
		MCDBR	8.0	8.8	10.0	11.2	11.3	11.3	11.0	10.5	9.7	9.6	8.5	7.9
	5% WB	MCWBR	4.9	5.2	5.8	6.3	6.3	6.3	6.0	5.9	5.7	5.2	4.6	4.5
		MCWBR	7.4	8.2	9.1	10.4	10.4	10.6	10.3	9.9	9.0	8.6	7.7	7.6
(40) Clear-Sky Solar Irradiance	taub		0.248	0.274	0.318	0.358	0.376	0.379	0.377	0.365	0.342	0.317	0.280	0.248
		taud	2.342	2.281	2.235	2.234	2.261	2.297	2.295	2.341	2.398	2.448	2.452	2.383
	Ebn at Noon		865	903	905	895	889	885	881	879	872	845	819	825
		Edh at Noon	72	95	117	130	131	127	126	115	99	81	66	62
(44) All-Sky Solar Radiation	RadAvg		1.40	2.28	3.41	4.38	4.97	5.89	5.82	5.07	3.98	2.59	1.47	1.14
	RadStd		0.16	0.32	0.48	0.62	0.43	0.55	0.50	0.46	0.35	0.27	0.20	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (112 neighbors)		N/A	N/A	N/A	+0.73	+0.41	+0.31	N/A	N/A	+57

Nomenclature: See separate page